

Fantasy Premier League

Table of contents

The Beautiful Game	3
Preseason	4
The Goal	4
The Goalposts	4
Relegation zone	4
Promotion zone	4
Safe zone	5
Formation	5
Financial fairplay	5
Pitch limits	5
GW 1 - The Initial Squad	6
Sample FPL API Data	6
Players	6
Teams	6
Positions	7
Combined data	7
Prior Season Data	8
The Initial Squad	8
Goalkeepers	9
Defenders	10
Midfielders	10
Forwards	11
Leftover Capital	12

The Beautiful Game

This is my journey of playing FPL with 0 ball knowledge and a bit of Python knowledge.

Preseason

The Goal

The purpose of this project is to see how well I can perform using a systematic set of trading rules based on historical data, rather than any knowledge or experience of football

The Goalposts

What does “performing well” mean? The most natural measure of performance is percentile rank among all FPL players, but that doesn’t have an objective benchmark. Is 50% good? 33%?

Perhaps the most objective measure of my performance is the difference between my final point score and the score I would have ended on if I had made no transfers throughout the season. This will evaluate the quality of my transfer decisions, but not of my initial choice of squad. I think I am happy with this, because I expect my initial squad to be far from both the best and worst initial squad.

Since my evaluation method is essentially controlling for the initial squad choice, once the season is over I might be able to work out how well my initial squad would have done with 0 transfers, compared to all the other possibilities.

Relegation zone

That leaves me with a baseline target of ensuring my final points is higher than the points my initial squad would have ended up on. With 37 transfers

Promotion zone

A target that feels certainly ambitious and objective (although, impossible to gauge if at all realistic) target might be to end on more than 1.5x the points that my initial squad would have

Safe zone

Ending up in the Relegation Zone will be an embarrassing but fun story to tell. Ending up in the Promotion Zone will be an even more fun story to tell. If both elude me, then at least I will have the code for my CV.

Formation

How should I choose when to buy, sell, and hold players? And on the basis of what data? I will be using moving averages to make my decisions. I will aim to buy players who are perform well over the last 3 weeks, and vice versa for selling. I will not waste my transfers transacting on averagely performing players, meaning I might not buy good players who are not good enough, and might not sell (i.e. will hold) players who are not (yet) performing badly enough.

Financial fairplay

I will only be using data available from the official FPL API. More specifically, I'll only be using data pertaining to players' points scored in the current and past two seasons. I will not make any decisions based on qualitative or other quantitative data I receive in any form or via any means, external to the API. Secondly, this means I am not "cheating". Primarily, this is because I have no reason to believe I'll be able to employ this information better than that from the API.

Pitch limits

Currently undecided if I want to use chips. I think if I do, I will resign myself to using them the week before they expire, or not at all, on the basis that if I am going to use them, I should use them when I have the most information to hand. But, I don't think I'll use them, because it'll make it very difficult to compare my performance with my initial squad's. I would have to work out how I would have used them, had I kept my initial squad, which feels like a lot of work.

GW 1 - The Initial Squad

With 0 ball knowledge, all I can go off when picking my initial squad is how well they performed last season, and perhaps the season before.

Sample FPL API Data

First, a look at the data we have available from the FPL API.

Players

	PID	first_name	second_name	now_cost
410	337	Brenden	Aaronson	53
579	477	Zach	Abbott	40
52	56	Tammy	Abraham	55
174	151	Josh	Acheampong	45
81	73	Tyler	Adams	50
598	147	Tosin	Adarabioyo	44
655	546	Simon	Adingra	49
85	77	Amine	Adli	49
319	650	David	Affengruber	45
263	633	Honest	Ahanor	45

Teams

	name	played	points	position
0	Arsenal	0	0	2
1	Aston Villa	0	0	16
2	Bournemouth	0	0	17
3	Brentford	0	0	4
4	Brighton	0	0	3
5	Chelsea	0	0	10

	name	played	points	position
6	Coventry City	0	0	18
7	Crystal Palace	0	0	15
8	Everton	0	0	7
9	Fulham	0	0	19

Positions

	singular_name	squad_max_play
0	Goalkeeper	1
1	Defender	5
2	Midfielder	5
3	Forward	3

Combined data

We'll need to combine this data to make use of it, and also declutter what I won't be using. To keep things simple, I'll only be using historical points scored to decide how to change my squad.

	PID	first_name	second_name	team_name	position_name	now_cost	total_points
136	124	Pascal	Groß	Brighton	Midfielder	58	47
272	229	James	Tarkowski	Everton	Defender	61	43
403	330	Jayden	Bogle	Leeds	Defender	46	42
491	411	Erling	Haaland	Man City	Forward	156	39
105	94	Kevin	Schade	Brentford	Midfielder	62	39
127	115	Maxim	De Cuyper	Brighton	Defender	49	38
474	391	Joško	Gvardiol	Man City	Defender	57	37
482	399	Rayan	Cherki	Man City	Midfielder	78	34
351	572	Konstantinos	Tzolakis	Hull City	Goalkeeper	46	34
459	379	Alexander	Isak	Liverpool	Forward	91	33
480	397	Antoine	Semenyo	Man City	Midfielder	84	33
184	165	João Pedro	Junqueira de Jesus	Chelsea	Forward	77	33
11	12	Bukayo	Saka	Arsenal	Midfielder	95	32
151	504	Luka	Vušković	Brighton	Defender	50	32
447	367	Cody	Gakpo	Liverpool	Midfielder	72	32
334	286	Mohamed	Belloumi	Hull City	Midfielder	51	31
517	426	Bruno	Borges Fernandes	Man Utd	Midfielder	119	31

	PID	first_name	second_name	team_name	position_name	now_cost	total_points
76	68	Marcus	Tavernier	Bournemouth	Midfielder	61	31
541	449	Lewis	Hall	Newcastle	Defender	53	31
471	388	Marc	Guéhi	Man City	Defender	60	30

Prior Season Data

The ‘total_points’ column will update each week. What I need for my initial squad selection are players’ total points from the last 2 seasons.

	PID	first_name	second_name	position_name	now_cost	points_2yr	points_per_cost
0	1	David	Raya Martín	Goalkeeper	61	304	4.983607
1	2	Kepa	Arrizabalaga Revuelta	Goalkeeper	49	108	2.204082
2	3	Illan	Meslier	Goalkeeper	49	0	0.000000
3	4	Gabriel	dos Santos Magalhães	Defender	80	326	4.075000
4	5	Jurriën	Timber	Defender	65	263	4.046154
5	6	William	Saliba	Defender	59	267	4.525424
6	7	Myles	Lewis-Skelly	Midfielder	54	66	1.222222
7	8	Riccardo	Calafiori	Defender	58	148	2.551724
8	9	Piero	Hincapié	Defender	52	87	1.673077
9	10	Benjamin	White	Defender	55	95	1.727273
10	11	Cristhian	Mosquera	Defender	53	40	0.754717
11	12	Bukayo	Saka	Midfielder	95	284	2.989474
12	13	Declan	Rice	Midfielder	74	311	4.202703
13	14	Eberechi	Eze	Midfielder	63	261	4.142857
14	15	Martin	Ødegaard	Midfielder	68	190	2.794118
15	16	Noni	Madueke	Midfielder	62	192	3.096774
16	17	Mikel	Merino Zazón	Midfielder	58	159	2.741379
17	18	Gabriel	Martinelli Silva	Midfielder	63	182	2.888889
18	19	Martín	Zubimendi Ibáñez	Midfielder	53	133	2.509434
19	20	Max	Dowman	Midfielder	53	16	0.301887

The Initial Squad

Before I choose which players I want, how do I choose how many of each position I want, subject to the maximum constraints? I’ll decide based on the average points scored for each position, over the last season:

Mean Goalkeeper Points over last season: 72
 Mean Defender Points over last season: 74
 Mean Midfielder Points over last season: 70
 Mean Forward Points over last season: 70
 Mean Goalkeeper cost: 4.8
 Mean Defender cost: 4.8
 Mean Midfielder cost: 5.6
 Mean Forward cost: 6.1

So, very crudely, Defenders are a bit more lucrative than other positions, and the opposite for Forwards. So I'll pick two Goalkeepers, then as many Defenders as I can, then as many Midfielders as I can, and then as few Forwards as I can. Perhaps this formation will change as the season progresses, depending on whether relative points/cost changes between different positions. I also will use as a tiebreaker a preference to add players from teams not already in my squad, to hopefully reduce the variance of my weekly points.

Below, we'll order players by the ratio of their points over the last two seasons, divided by their current cost. I'll then select for my initial squad players with a high ratio and a low cost. The idea is that I'm happy to buy underpriced players who score fewer points, so that I have the flexibility later in the season to use current season data to make transfers. Currently, I'm picking based on prior season performance, which it feels fair to assume is uniformly a worse indicator for future performance than current season data.

As we only have current cost, which is a dynamic field, the tables below are likely to show different players than the ones I chose at the time of writing. But, the logic is the same.

Goalkeepers

	PID	first_name	second_name	position_name	now_cost	points_2yr	points_per_cost_2yr
269	226	Jordan	Pickford	Goalkeeper	55	293	5.327273
233	198	Dean	Henderson	Goalkeeper	50	266	5.320000
292	250	Bernd	Leno	Goalkeeper	45	237	5.266667
123	109	Bart	Verbruggen	Goalkeeper	45	233	5.177778
569	467	Matz	Sels	Goalkeeper	50	255	5.100000
0	1	David	Raya Martín	Goalkeeper	61	304	4.983607
164	140	Robert	Lynch Sánchez	Goalkeeper	49	239	4.877551
160	28	Emiliano	Martínez Romero	Goalkeeper	50	231	4.620000
534	442	Nick	Pope	Goalkeeper	49	198	4.040816
94	82	Caoimhín	Kelleher	Goalkeeper	50	188	3.760000

I choose Verbruggen and Leno

Defenders

	PID	first_name	second_name	position_name	now_cost	points_2yr	points_per_cost_2yr
237	204	Tyrick	Mitchell	Defender	45	258	5.733333
568	201	Daniel	Muñoz Mejía	Defender	54	278	5.148148
471	388	Marc	Guéhi	Defender	60	295	4.916667
436	356	Virgil	van Dijk	Defender	65	318	4.892308
573	471	Nikola	Milenković	Defender	54	264	4.888889
597	112	Jan Paul	van Hecke	Defender	49	235	4.795918
96	84	Nathan	Collins	Defender	54	256	4.740741
571	469	Neco	Williams	Defender	50	237	4.740000
276	233	Vitalii	Mykolenko	Defender	46	216	4.695652
188	200	Maxence	Lacroix	Defender	59	268	4.542373
5	6	William	Saliba	Defender	59	267	4.525424
272	229	James	Tarkowski	Defender	61	275	4.508197
575	473	Ola	Aina	Defender	44	195	4.431818
26	31	Ezri	Konsa Ngoyo	Defender	46	203	4.413043
31	32	Matty	Cash	Defender	44	193	4.386364
540	448	Dan	Burn	Defender	48	208	4.333333
508	417	Diogo	Dalot Teixeira	Defender	49	206	4.204082
167	143	Trevoh	Chalobah	Defender	55	229	4.163636
30	30	Lucas	Digne	Defender	45	186	4.133333
60	611	Aaron	Wan-Bissaka	Defender	45	185	4.111111

I choose Mitchell, Guehi, Van Hecke, Tarkowski, and N.Williams

Midfielders

	PID	first_name	second_name	position_name	now_cost	points_2yr	points_per_cost_2yr
518	427	Bryan	Mbeumo	Midfielder	79	384	4.860759
301	261	Alex	Iwobi	Midfielder	54	259	4.796296
492	481	Elliot	Anderson	Midfielder	63	287	4.555556
107	96	Mikkel	Damsgaard	Midfielder	54	244	4.518519
105	94	Kevin	Schade	Midfielder	62	274	4.419355
480	397	Antoine	Semenyo	Midfielder	84	367	4.369048
448	368	Dominik	Szoboszlai	Midfielder	70	303	4.328571
27	452	Bruno	Guimarães Rodriguez Moura	Midfielder	67	289	4.313433
241	208	Ismaila	Sarr	Midfielder	62	266	4.290323
161	40	Morgan	Rogers	Midfielder	77	330	4.285714

	PID	first_name	second_name	position_name	now_cost	points_2yr	points_per
582	480	Morgan	Gibbs-White	Midfielder	80	342	4.275000
452	372	Alexis	Mac Allister	Midfielder	54	230	4.259259
466	155	Enzo	Fernández	Midfielder	69	292	4.231884
12	13	Declan	Rice	Midfielder	74	311	4.202703
467	237	Iliman	Ndiaye	Midfielder	58	242	4.172414
13	14	Eberechi	Eze	Midfielder	63	261	4.142857
547	457	Jacob	Murphy	Midfielder	59	241	4.084746
451	371	Ryan	Gravenberch	Midfielder	58	236	4.068966
519	428	Matheus	Santos Carneiro da Cunha	Midfielder	79	321	4.063291
631	525	Mateus	Fernandes	Midfielder	57	230	4.035088

I choose Szoboszlai, Rice, Anderson, Iwobi, and Damsgaard

Forwards

	PID	first_name	second_name	position_name	now_cost	points_2yr	points_per
51	55	Ollie	Watkins	Forward	78	353	4.525641
163	136	Danny	Welbeck	Forward	59	251	4.254233
255	223	Jean-Philippe	Mateta	Forward	63	264	4.190476
591	490	Chris	Wood	Forward	58	241	4.155172
87	79	Francisco Evanilson	de Lima Barbosa	Forward	60	243	4.050000
184	165	João Pedro	Junqueira de Jesus	Forward	77	303	3.935063
254	222	Jørgen	Strand Larsen	Forward	60	220	3.666667
288	248	Norberto Bercique	Gomes Betuncal	Forward	54	191	3.537037
137	125	Georginio	Rutter	Forward	53	182	3.433962
554	464	Yoane	Wissa	Forward	62	212	3.419355
418	346	Dominic	Calvert-Lewin	Forward	60	205	3.416667
567	167	Liam	Delap	Forward	56	177	3.160714
459	379	Alexander	Isak	Forward	91	252	2.769231
491	411	Erling	Haaland	Forward	156	420	2.692308
633	527	Richarlison	de Andrade	Forward	59	154	2.610169
632	526	Dominic	Solanke-Mitchell	Forward	58	148	2.551724
115	106	Igor Thiago	Nascimento Rodrigues	Forward	78	190	2.435897
310	271	Rodrigo	Muniz Carvalho	Forward	54	122	2.259259
531	440	Joshua	Zirkzee	Forward	53	106	2.000000
592	491	Igor Jesus	Maciel da Cruz	Forward	58	114	1.965517

I choose Calvert-Lewin, Mateta, and Welbeck

Leftover Capital

This leaves me with £14m to play with. If I make 38 transfers over the season, I can average a net cost of ~£0.5m per transfer. The idea is that every transfer I make will use some capital to improve my team made from old data, on the basis of new data (current season data). I expect my initial squad to optimal with respect to the old data (as far as my selection logic is concerned), but to be a very poor predictor of this season's performance when compared with current season data, as it eventually emerges. Selecting a cheap initial squad gives me flexibility to leverage current season data to make "corrections" to my initial squad that was chosen from old data.